

SET AND RELATION

✓ Set: A unordered collection of distinct elements of same type.

eg $A = \{1, 2, 3, 4\}$

↳ set of +ive integers

$B = \{\text{red, white, Pink, Blue}\}$

→ Uses curly braces $\{ \}$

→ denoted by Capital symbol

Cardinality of Set: # of elements in a set.

eg. if $A = \{1, 2, 3, 4\}$

then $|A| = 4$.

Null set: $\{ \}$ or ϕ

empty / no element.

and if $A = \{ \}$, then $|A| = 0$ or $A = \phi$.

✓ Power set: of a set is a set of every possible subset.

eg. $A = \{1, 2, 3\}$, then

$P(A) = \{ \phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\} \}$

If $|A| = n$, then $|P(A)| = 2^n$

if $|A| = 3$, then $|P(A)| = 2^3 = 8$

✓ Subset: $A \subseteq B$, A is a subset of B.

All the elements which belong to A, also belongs to B.

eg. $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$
 so, $A \subseteq B$

Also, if $A = \{1, 2, 3, 4\}$ then $A \subseteq B$.
Proper Subset
 $A \subset B$

$\Rightarrow A \neq B$

Relation:

✓ Any possible subset of a cartesian product.

$A = \{1, 2\}$, $B = \{a, b, c\}$

$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$

- First element from the first set and second element from the second set.

- Ordered set of pairs.

- Any possible subset of this cartesian product will be a relation.

eg. $R = \{(1, a), (1, b), (2, a)\}$

- maximum relation possible $\Rightarrow A \times B$.

Min $\Rightarrow \phi$.

1. $\{(1, a)\}$
2. $\{(1, a), (1, b)\}$
3. $\{(1, a), (1, b), (1, c)\}$
4. $\{(1, a), (1, b), (1, c), (2, a)\}$
5. $\{(1, a), (1, b), (1, c), (2, a), (2, b)\}$
6. $\{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$

I have selected only 6, but there are $2^{3 \cdot 2}$ such subsets and the max reln possible is all the subsets that is ⑥

→ If $|A| = m, |B| = n$

Total # of relation possible $\Rightarrow \boxed{2^{m \cdot n}}$

Type of relation

① Reflexive reln: $A \times A \quad \forall a \in A, (a, a) \in R$

↓
diagonal elements

eg. if

	1	2	3
1	11	12	13
2	21	22	23
3	31	32	33

Diagonal pair-wise is the minimum condition, i.e. $A = \{(1,1), (2,2), (3,3)\}$

eg. $B = \{(1,1), (2,2), (1,3)\}$ it is not a reflexive reln. since $(3,3)$ is not present. Min condition is that all the diagonal elements should be present.

→ If each element of set X is related or linked to itself then a binary relation across a set X is reflexive.

eg. consider a set $A = \{1, 2\}$, Now the reflexive relation will be $R = \{(1,1), (2,2), (1,2), (2,1)\}$. Hence $(a,a) \in R \quad \forall a \in A$ where a is the element, A is the set and R is the relation.

→ If A set has n elements.

Then the total reflexive Relation is $\boxed{2^{n^2-n}}$

How does the formula work.

A relation R is reflexive if the matrix diagonal elements are 1.

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$$

If we take a closer look the matrix, we can notice that the size of matrix is n^2 . The n diagonal entries are fixed. For remaining $n^2 - n$ entries, we have to either fill 0 or 1.

So there are 2^{n^2-n} ways of filling the matrix.

Anti reflexivity /

② Irrreflexive Relation: $\forall a \in A$ and $(a, a) \notin R$

Diagonal pairs are not allowed.

eg, $A = \{(1,1), (2,1), (3,1)\}$

③ Symmetric Relation: $\forall a, b \in A$ if $(a, b) \in R$ then $(b, a) \in R$

eg,

$$A = \{(1,2), (2,1)\}$$

$$A1 = \{(1,2), (1,3), (3,1), (2,1)\}$$

$$A = \begin{bmatrix} 11 & 12 & 13 \\ 21 & 22 & 23 \\ 31 & 32 & 33 \end{bmatrix}$$

Hence $A1$ is a symmetric relation.

$$A2 = \{(1,1), (1,2), (3,1), (1,3)\}$$

Not a symmetric relation.

④ Antisymmetric : If $(a,b) \in R$
 $(b,a) \in R$
 then $(a=b)$.

OR

If $(a,b) \in R$
 then $(b,a) \notin R$
 but diagonal pairs are
 allowed.

eg $(1,1)$ allowed but if $(1,2)$ present
 then $(2,1)$ should not be present.

① If $A = \{(1,1), (1,2), (2,1)\}$ then this is
 not an Antisymmetric.

② $B = \{(1,1), (1,2), (2,2)\}$ is an Antisymmetric
 ic.

⑤ Asymmetric : If $(a,b) \in R$ then
 $(b,a) \notin R$
 and diagonal pairs are not
 allowed.

eg.



$A = \{1, 2, 3\}$ $B = \{2, 3\}$, $C = \{2, 3\}$
 NO NO YES

⑥ Transitive Relation:

$$\forall a, b, c \in A$$

$$\text{If } (a, b) \in R$$

$$(b, c) \in R$$

$$\text{then } \underline{(a, c) \in R}$$

eg. $A = \{(1, 2), (2, 1), (2, 3), (1, 3)\}$



No

Since, $(1, 2) \in R$ and $(2, 1) \in R$

but $(1, 1)$ not present in the set.

$A = \{(1, 2), (2, 1), (2, 3), (1, 3), (1, 1), (2, 2)\}$
is a transitive relation.

Properties of the Comparison Relations

① Equal to ($=$)

$$\text{then } \Rightarrow$$

Reflexivity: $a = a$

Symmetry: $a = b \Rightarrow b = a$

Transitivity: $a = b, b = c, \text{ then } a = c$

② Not equal to ($\neq$)

Antireflexivity: not $a \neq a$

Symmetry: $a \neq b \Rightarrow b \neq a$

Not reflexive since for no number a
 $a \neq a$.

③ Less than ($<$)

Antireflexivity: not $a < a$

Asymmetry: $a < b \Rightarrow \text{not } b < a$

Transitivity: $a < b, b < c \Rightarrow a < c$

④ Greater than ($>$)

Anti-reflexivity: not $a > a$

Asymmetry: $a > b \Rightarrow \text{not } b > a$

Transitivity: $a > b, b > c \Rightarrow a > c$

⑤ Less than or equal to ($\leq$)

Reflexivity: $a \leq a$

Antisymmetry: $a \leq b, b \leq a \Rightarrow a = b$

Transitivity: $a \leq b, b \leq c \Rightarrow a \leq c$

⑥ Greater than or equal to ($\geq$)

Same as ⑤

Defn

An Equivalence relation is reflexive, symmetric and transitive.